5.3.7 Wake-up time from low-power modes and voltage scaling transition times

The wake-up times given in the table below are the latency between the event and the execution of the first user instruction.

The device goes in low-power mode after the WFE (wait for event) instruction

Table 30. Wake-up time from low-power modes

1. Evaluated by characterization - Not tested in production.
2. The wakeup times are measured from the wakeup event to the point in which the application code reads the first instruction.

Symbol	Parameter	Conditions	Wakeup clock	f_{HCLK} (MHz)	Typ	Max	Unit
$t_{wu(Sleep)}$	Wakeup from Sleep	Instruction cache enabled or disabled	-	-	16		CPU clock cycles
$t_{wu(Stop)}$	Wakeup from Stop 0	Flash memory in normal mode	HSI	144	3.6	5	µs
		Flash memory in low-power mode	HSI	144	7.2	11	
		Flash memory in normal mode	HSIDIV3	48	5.1	7	
		Flash memory in low-power mode	HSIDIV3	48	8.6	13.5	
	Wakeup from Stop 1[1]	Flash memory in normal mode	HSI	144	37.0	49	
		Flash memory in low-power mode	HSI	144	40.5	58	
		Flash memory in normal mode	HSIDIV3	48	39.0	53	
		Flash memory in low-power mode	HSIDIV3	48	42.0	61	
$t_{wu(Standby)}$	Wakeup from Standby mode[1]	-	HSIDIV3	48	445	-	

1. *Those parameters depend on V_{CAP} capacitance value and V_{CAP} voltage at the instant of the wake-up event.*

Table 31. Wake-up time using USART/LPUART

Symbol	Parameter	Condition	Typ	Max[1]	Unit
$t_{WUUSART}$/ $t_{WULPUART}$	Wake-up time needed to calculate the maximum USART/LPUART baudrate allowing to wake up from Stop mode when USART/LPUART clock source is 48 MHz by HSIDIV3	Stop 0 mode	4.8	6.6	µs
		Stop 1 mode			

1. *Specified by design - Not tested in production.*

5.3.8 External clock timing characteristics

5.3.8.1 *High-speed external user clock generated from an external source*

In bypass mode, the HSE oscillator is switched off and the input pin is a standard GPIO.

The external clock signal has to respect the I/O characteristics in I/O port characteristics. However, the recommended clock input waveform is shown in the figure below.

Table 32. High-speed external user clock characteristics

Specified by design and not tested in production.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{HSE_ext}	User external clock source frequency	Digital mode (HSEYBYP = 1, HSEEXT = 1)	-	-	50	MHz
		Analolog mode (HSEYBYP = 1, HSEEXT = 0)	4	-	50	
V_{HSEH}	OSC_IN input pin high-level voltage	Digital mode (HSEYBYP = 1, HSEEXT = 1)	$0.7 \times V_{DD}$	-	V_{DD}	V
V_{HSEL}	OSC_IN input pin low-level voltage	Digital mode (HSEYBYP = 1, HSEEXT = 1)	V_{SS}	-	$0.3 \times V_{DD}$	
$t_{w(HSEH)}$[1] $t_{w(HSEL)}$[1]	OSC_IN high or low time	Digital mode (HSEYBYP = 1, HSEEXT = 1)	7	-	-	ns
$DuCy_{HSE}$	OSC_IN duty cycle	Digital mode (HSEYBYP = 1, HSEEXT = 1)	45	-	55	%